21 forecasters on the same test windows of nine populations. a, b: means over units, so larger units weigh more; c, d: the median unit. Bold: lowest in the column. Rows in the order of figure 2. ankyra-lite 0.2.0: carrier recipe Rw (trained from scratch with early stopping on the 93-day validation set, then 8 more epochs with the validation span returned and each training window's loss weighted by the inverse-noise rule; six seeds averaged; figure 1b), the anchoring layer, the state rule and the origin-continuity layer; hourly errors score the delivered trajectory after that layer, energy, level and peak are read before it (readout rule A). Linear-context: the TiDE implementation of the release, whose forecast is a linear map of the load context.

**a RMSE (kW, mean over units)**

| | BDG2 | Cambridge | HEEW | EWELD | GoiEner NH | GoiEner HH | Oslo | Drammen | CINELDI |
|---|---|---|---|---|---|---|---|---|---|
| **ankyra-lite** | **28.8** | **12.7** | 26.1 | 98.8 | **0.550** | **0.170** | **22.5** | **13.3** | **16.3** |
| Ridge (per unit) | 30.1 | 15.0 | 29.3 | 103 | 0.590 | 0.173 | 28.9 | 14.7 | 20.6 |
| Chronos-2-X | 30.1 | 14.5 | 27.3 | 106 | 0.562 | 0.177 | 26.2 | 14.4 | 19.0 |
| iTransformer-X | 32.9 | 13.8 | 31.1 | 103 | 0.641 | 0.180 | 25.2 | 14.9 | 18.2 |
| iTransformer | 33.2 | 14.5 | 30.8 | 105 | 0.633 | 0.183 | 26.9 | 15.3 | 17.2 |
| PatchTST | 32.8 | 14.7 | 28.3 | 104 | 0.651 | 0.178 | 26.5 | 15.2 | 17.2 |
| GBT-T | 30.6 | 17.6 | **25.5** | 154 | 0.809 | 0.183 | 23.1 | 14.8 | 16.7 |
| TimesFM-X | 32.0 | 15.8 | 27.7 | 102 | 0.606 | 0.176 | 29.0 | 15.8 | 21.1 |
| TimesFM | 31.2 | 15.6 | 27.6 | 100 | 0.643 | 0.177 | 29.1 | 17.0 | 21.4 |
| Chronos-2 | 31.9 | 15.9 | 29.6 | **97.1** | 0.637 | 0.177 | 29.1 | 16.4 | 18.5 |
| DLinear | 30.7 | 16.8 | 32.2 | 121 | 0.640 | 0.180 | 29.5 | 15.4 | 18.6 |
| Linear-context | 30.7 | 17.4 | 32.3 | 122 | 0.644 | 0.180 | 29.6 | 15.4 | 18.4 |
| 4-week profile | 34.8 | 15.2 | 38.2 | 111 | 0.729 | 0.188 | 31.6 | 18.9 | 18.8 |
| Holt–Winters | 37.0 | 16.9 | 33.1 | 115 | 0.723 | 0.196 | 33.6 | 18.4 | 27.1 |
| LSTM | 32.9 | 27.9 | 39.7 | 144 | 0.692 | 0.182 | 39.4 | 18.3 | 22.6 |
| Previous-month profile | 36.9 | 16.7 | 39.9 | 110 | 0.775 | 0.185 | 36.4 | 20.4 | 21.6 |
| MSTL | 35.8 | 17.8 | 30.6 | 109 | 0.731 | 0.213 | 34.9 | 18.7 | 26.0 |
| Seasonal naive (week) | 36.2 | 17.9 | 33.2 | 115 | 0.727 | 0.214 | 34.5 | 18.6 | 24.0 |
| Last-year profile | 37.6 | 21.6 | 38.8 | 249 | 0.698 | 0.246 | 33.1 | 18.8 | 21.5 |
| GBT (zero-shot) | 52.0 | 18.2 | 41.2 | 135 | 0.846 | 0.231 | 44.3 | 20.5 | 24.9 |
| Seasonal naive (day) | 40.9 | 22.6 | 36.1 | 106 | 0.826 | 0.216 | 47.0 | 24.8 | 31.9 |

**b MAE (kW, mean over units)**

| | BDG2 | Cambridge | HEEW | EWELD | GoiEner NH | GoiEner HH | Oslo | Drammen | CINELDI |
|---|---|---|---|---|---|---|---|---|---|
| **ankyra-lite** | 20.4 | **7.80** | 19.3 | 66.0 | 0.343 | 0.100 | **15.2** | **8.96** | 11.5 |
| Ridge (per unit) | 21.5 | 9.33 | 21.0 | 66.9 | 0.362 | 0.104 | 19.9 | 10.1 | 13.7 |
| Chronos-2-X | **20.4** | 8.76 | 19.1 | 67.2 | **0.311** | **0.0927** | 16.7 | 9.62 | 10.5 |
| iTransformer-X | 23.5 | 8.56 | 22.2 | 69.7 | 0.403 | 0.109 | 16.8 | 10.2 | 12.5 |
| iTransformer | 24.0 | 9.01 | 22.1 | 70.1 | 0.398 | 0.110 | 17.8 | 10.3 | 11.9 |
| PatchTST | 23.8 | 9.22 | 20.7 | 70.7 | 0.418 | 0.106 | 17.9 | 10.5 | 12.5 |
| GBT-T | 21.9 | 9.96 | **18.4** | 90.8 | 0.489 | 0.114 | 15.7 | 10.2 | 11.4 |
| TimesFM-X | 22.6 | 9.76 | 19.3 | 65.1 | 0.358 | 0.0963 | 19.8 | 11.0 | 13.3 |
| TimesFM | 21.4 | 9.35 | 19.0 | 63.4 | 0.362 | 0.0943 | 18.9 | 11.3 | 12.7 |
| Chronos-2 | 21.9 | 9.56 | 20.2 | **60.7** | 0.353 | 0.0927 | 18.2 | 10.7 | **10.0** |
| DLinear | 22.4 | 11.3 | 24.6 | 93.7 | 0.418 | 0.112 | 20.4 | 10.5 | 12.8 |
| Linear-context | 22.4 | 11.9 | 24.6 | 94.8 | 0.425 | 0.111 | 20.6 | 10.6 | 12.6 |
| 4-week profile | 25.1 | 9.19 | 27.3 | 72.8 | 0.430 | 0.109 | 21.0 | 13.5 | 11.6 |
| Holt–Winters | 26.8 | 10.7 | 24.2 | 78.4 | 0.447 | 0.126 | 22.4 | 12.9 | 18.8 |
| LSTM | 24.6 | 21.8 | 30.5 | 118 | 0.467 | 0.115 | 29.3 | 12.6 | 17.6 |
| Previous-month profile | 28.2 | 11.5 | 29.5 | 73.9 | 0.521 | 0.112 | 28.1 | 15.3 | 16.9 |
| MSTL | 25.2 | 11.2 | 21.6 | 71.0 | 0.449 | 0.135 | 23.2 | 12.6 | 17.1 |
| Seasonal naive (week) | 24.4 | 11.0 | 22.3 | 72.5 | 0.399 | 0.117 | 21.7 | 12.4 | 13.9 |
| Last-year profile | 26.0 | 14.5 | 28.5 | 196 | 0.409 | 0.139 | 22.0 | 12.9 | 13.0 |
| GBT (zero-shot) | 30.8 | 11.8 | 27.5 | 84.9 | 0.519 | 0.139 | 27.8 | 14.0 | 17.9 |
| Seasonal naive (day) | 27.9 | 14.5 | 24.9 | 66.6 | 0.477 | 0.122 | 30.3 | 16.3 | 20.9 |

**c CV(RMSE) (%, median unit)**

| | BDG2 | Cambridge | HEEW | EWELD | GoiEner NH | GoiEner HH | Oslo | Drammen | CINELDI |
|---|---|---|---|---|---|---|---|---|---|
| **ankyra-lite** | 19.7 | **16.1** | **16.3** | 53.9 | **69.1** | **96.9** | **28.2** | **29.3** | **38.6** |
| Ridge (per unit) | 22.6 | 19.4 | 17.9 | 55.3 | 71.1 | 98.6 | 36.1 | 32.7 | 43.2 |
| Chronos-2-X | **18.1** | 18.8 | 17.0 | 52.5 | 78.0 | 102.6 | 31.8 | 32.2 | 44.4 |
| iTransformer-X | 23.2 | 17.6 | 18.3 | 55.1 | 74.8 | 101.0 | 31.9 | 32.8 | 47.3 |
| iTransformer | 22.9 | 19.5 | 19.3 | 57.7 | 76.2 | 103.0 | 33.0 | 32.4 | 45.0 |
| PatchTST | 23.4 | 19.3 | 19.8 | 55.8 | 78.8 | 101.9 | 33.0 | 33.2 | 42.9 |
| GBT-T | 19.5 | 17.7 | 18.6 | 68.6 | 89.3 | 102.9 | 30.4 | 33.6 | 46.4 |
| TimesFM-X | 23.0 | 20.9 | 18.1 | 56.3 | 78.0 | 101.7 | 37.5 | 35.4 | 44.0 |
| TimesFM | 20.9 | 18.7 | 17.5 | 55.9 | 77.4 | 102.9 | 37.5 | 37.1 | 43.7 |
| Chronos-2 | 21.9 | 20.4 | 18.5 | **51.9** | 76.8 | 102.4 | 36.8 | 35.5 | 41.5 |
| DLinear | 23.0 | 22.7 | 19.7 | 56.8 | 80.3 | 102.1 | 37.1 | 34.9 | 43.7 |
| Linear-context | 23.3 | 23.9 | 19.8 | 56.7 | 81.0 | 102.0 | 37.7 | 35.1 | 44.0 |
| 4-week profile | 24.5 | 19.9 | 21.0 | 56.5 | 83.5 | 106.9 | 41.3 | 42.3 | 48.3 |
| Holt–Winters | 25.1 | 21.1 | 21.1 | 68.7 | 83.8 | 110.4 | 44.6 | 41.6 | 57.5 |
| LSTM | 25.0 | 29.3 | 25.3 | 65.8 | 85.0 | 102.4 | 42.9 | 42.8 | 56.1 |
| Previous-month profile | 25.8 | 23.5 | 22.0 | 57.4 | 82.3 | 104.6 | 45.9 | 48.5 | 52.5 |
| MSTL | 26.0 | 23.0 | 20.0 | 67.1 | 90.7 | 123.0 | 44.2 | 42.0 | 53.9 |
| Seasonal naive (week) | 25.1 | 23.3 | 20.5 | 67.1 | 93.8 | 125.0 | 43.5 | 42.8 | 51.1 |
| Last-year profile | 25.8 | 27.9 | 27.2 | 94.0 | 97.8 | 133.4 | 40.4 | 43.2 | 64.0 |
| GBT (zero-shot) | 34.4 | 24.4 | 25.5 | 74.1 | 94.1 | 120.8 | 48.3 | 48.3 | 58.7 |
| Seasonal naive (day) | 29.1 | 30.0 | 21.6 | 63.7 | 102.9 | 124.0 | 59.4 | 60.0 | 77.1 |

**d WAPE (%, median unit)**

| | BDG2 | Cambridge | HEEW | EWELD | GoiEner NH | GoiEner HH | Oslo | Drammen | CINELDI |
|---|---|---|---|---|---|---|---|---|---|
| **ankyra-lite** | 13.9 | **11.3** | 11.6 | 34.1 | 40.7 | 56.5 | **19.1** | **19.6** | 25.8 |
| Ridge (per unit) | 15.8 | 13.1 | 12.2 | 34.1 | 43.6 | 58.8 | 25.7 | 22.6 | 28.2 |
| Chronos-2-X | **12.6** | 11.8 | **10.8** | 32.6 | 39.9 | 52.3 | 20.5 | 21.3 | **25.3** |
| iTransformer-X | 16.7 | 12.7 | 13.3 | 35.5 | 45.4 | 60.4 | 20.6 | 22.8 | 33.0 |
| iTransformer | 17.1 | 13.7 | 13.3 | 36.9 | 44.3 | 61.0 | 21.8 | 21.6 | 28.7 |
| PatchTST | 17.2 | 13.6 | 13.9 | 38.3 | 48.9 | 59.5 | 21.6 | 22.9 | 29.3 |
| GBT-T | 14.6 | 12.8 | 13.0 | 42.4 | 50.7 | 61.5 | 20.7 | 22.8 | 31.0 |
| TimesFM-X | 15.8 | 12.9 | 12.2 | 33.6 | 43.2 | 54.4 | 25.2 | 25.3 | 28.2 |
| TimesFM | 14.4 | 12.3 | 11.3 | 32.7 | 40.5 | 54.0 | 23.9 | 25.2 | 27.4 |
| Chronos-2 | 15.5 | 12.6 | 11.6 | **31.2** | **39.7** | **52.0** | 24.0 | 23.1 | 26.6 |
| DLinear | 16.6 | 16.4 | 14.7 | 41.2 | 49.7 | 62.3 | 25.8 | 23.6 | 32.3 |
| Linear-context | 16.6 | 17.2 | 14.6 | 41.9 | 50.7 | 62.2 | 25.9 | 23.8 | 32.9 |
| 4-week profile | 17.8 | 12.9 | 14.0 | 36.7 | 44.9 | 61.8 | 28.4 | 30.4 | 31.6 |
| Holt–Winters | 18.8 | 14.8 | 14.5 | 43.7 | 50.8 | 65.9 | 28.9 | 28.4 | 37.8 |
| LSTM | 18.5 | 18.9 | 18.3 | 49.4 | 52.5 | 63.0 | 30.7 | 29.0 | 41.7 |
| Previous-month profile | 20.5 | 16.7 | 16.4 | 38.0 | 54.3 | 62.9 | 35.0 | 36.4 | 39.3 |
| MSTL | 17.6 | 15.1 | 14.0 | 40.5 | 53.9 | 77.2 | 29.3 | 28.1 | 38.0 |
| Seasonal naive (week) | 16.4 | 14.5 | 13.6 | 38.0 | 48.6 | 68.6 | 28.1 | 27.9 | 31.4 |
| Last-year profile | 17.3 | 19.2 | 20.3 | 70.5 | 55.7 | 75.3 | 26.5 | 29.0 | 42.0 |
| GBT (zero-shot) | 22.2 | 16.2 | 17.2 | 44.8 | 57.9 | 72.3 | 34.0 | 32.6 | 40.2 |
| Seasonal naive (day) | 20.1 | 19.3 | 15.3 | 37.6 | 54.0 | 70.5 | 38.0 | 40.0 | 49.1 |

≤ −30 −15 0 +15 ≥ +30

Loss relative to ankyra-lite in the same column (%): red lower than ankyra-lite, blue higher

Same information · trained
Same information · zero-shot foundation model
Load only · zero-shot foundation model
Load only · statistical
Load only · trained
Per-unit regression (calendar, temperature)
Profiles, naive, zero-shot GBT